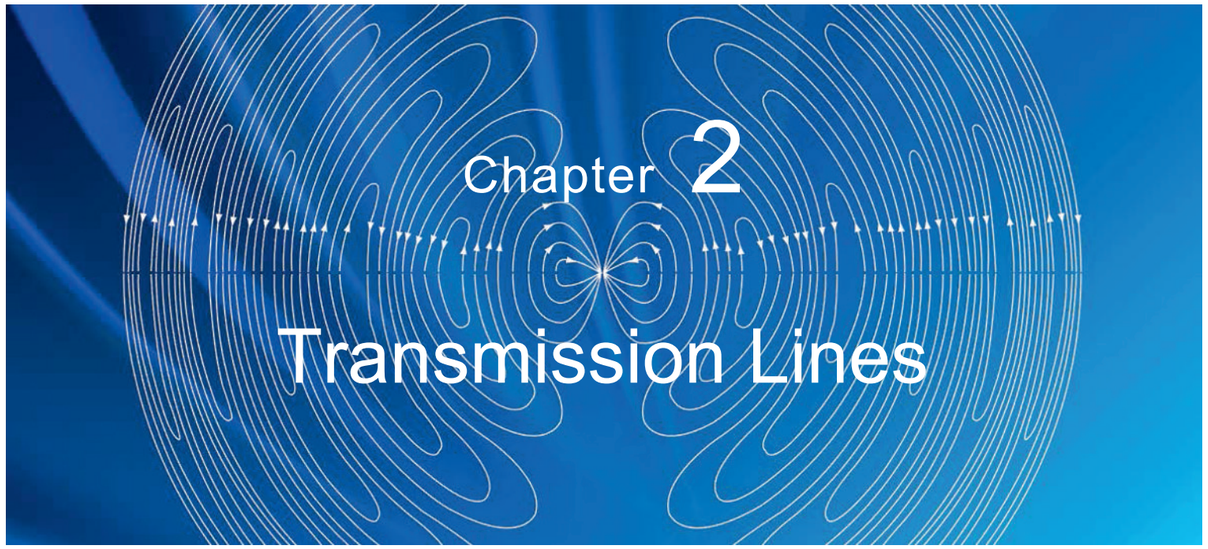


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Objectives

Upon learning the material presented in this chapter, you should be able to:

1. Calculate the line parameters, characteristic impedance, and propagation constant of coaxial, two-wire, parallel-plate, and microstrip transmission lines.
2. Determine the reflection coefficient at the load-end of the transmission line, the standing-wave pattern, and the locations of voltage and current maxima and minima.
3. Calculate the amount of power transferred from the generator to the load through the transmission line.
4. Use the Smith chart to perform transmission-line calculations.
5. Analyze the response of a transmission line to a voltage pulse.

第2章

传输线

章节内容

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目标

学习本章介绍的材料后，您应该能够：

1. 计算同轴、两线、平行板和微带传输线的线路参数、特性阻抗和传播常数。 2. 确定传输线负载端的反射系数、驻波模式以及电压和电流最大值和最小值的位置。 3. 计算通过传输线从发电机传输到负载的电量。 4. 使用史密斯圆图执行传输线计算。 5. 分析传输线对电压脉冲的响应。



Figure 2-1 A transmission line is a two-port network connecting a generator circuit at the sending end to a load at the receiving end.

2-1 General Considerations

In most electrical engineering curricula, the study of electromagnetics is preceded by one or more courses on electrical circuits. In this book, we use this background to build a bridge between circuit theory and electromagnetic theory. The bridge is provided by transmission lines, which is the topic of this chapter. By modeling transmission lines in the form of equivalent circuits, we can use Kirchhoff's voltage and current laws to develop wave equations whose solutions provide an understanding of wave propagation, standing waves, and power transfer. Familiarity with these concepts facilitates the presentation of material in later chapters.

Although the notion of ***transmission lines*** may encompass all structures and media that serve to transfer energy or information between two points, including nerve fibers in the human body and fluids and solids that support the propagation of mechanical pressure waves, this chapter focuses on transmission lines that guide electromagnetic signals. Such transmission lines include telephone wires, coaxial cables carrying audio and video information to TV sets or digital data to computer monitors, microstrips printed on microwave circuit boards, and optical fibers carrying light waves for the transmission of data at very high rates.

Fundamentally, a transmission line is a two-port network, with each port consisting of two terminals, as illustrated in Fig. 2-1. One of the ports, the line's sending end, is connected to a source (also called the generator). The other port, the line's receiving end, is connected to a load. The source connected to the transmission line's sending end may be any circuit generating an output voltage, such as a radar transmitter, an amplifier, or a computer terminal operating in transmission mode. From circuit theory, a dc source can be represented by a Thévenin-equivalent ***generator circuit*** consisting of a generator voltage V_g in series with a generator resistance R_g, as shown in Fig. 2-1. In the case of alternating-current (ac) signals, the generator circuit is represented by a voltage phasor $\tilde{V}_g$ and an impedance Z_g.

The load circuit, or simply the ***load***, may be a transmitting antenna in the case of radar, a computer terminal operating in the receiving mode, the input terminals of an amplifier, or any output circuit whose input terminals can be represented by an equivalent load impedance Z_L.

2-1.1 The Role of Wavelength

In low-frequency circuits, circuit elements usually are interconnected using simple wires. In the circuit shown in Fig. 2-2, for example, the generator is connected to a simple RC load via a pair of wires. In view of our definition in the preceding paragraphs of what constitutes a transmission line, we pose the question: *Is the pair of wires between terminals AA′ and terminals BB′ a transmission line?* If so, under what set of circumstances should we explicitly treat the pair of wires as a transmission line as opposed to ignoring their presence altogether and treating the circuit as only an RC load connected to a generator $\tilde{V}_g$? The answer to the first question is: *Yes,*

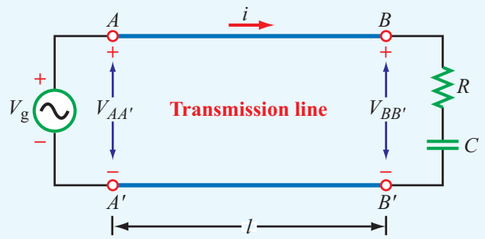


Figure 2-2 Generator connected to an RC circuit through a transmission line of length l.

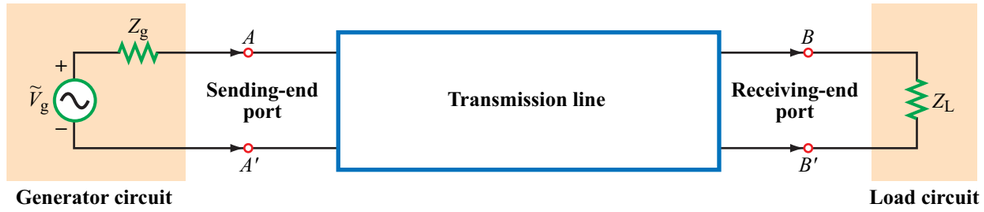


图2-1 A传输 在 线路是连接发送端发生器电路的二端口网络 ng 端到接收端的负载。

2-1 一般注意事项

在大多数电气工程课程中，电磁学学习之前都会有一门或多门电路课程。在本书中，我们利用这一背景搭建了电路理论和电磁理论之间的桥梁。电桥由传输线提供，这是本章的主题。通过以等效电路的形式对传输线进行建模，我们可以使用基尔霍夫电压和电流定律来建立波动方程，其解可以帮助我们理解波传播、驻波和功率传输。熟悉这些概念有助于在后面的章节中介绍材料。

尽管传输线的概念可能涵盖用于在两点之间传输能量或信息的所有结构和介质，包括人体中的神经纤维以及支持机械压力波传播的流体和固体，但本章重点讨论引导电磁信号的传输线。此类传输线包括电话线、将音频和视频信息传送到电视机或将数字数据传送到计算机显示器的同轴电缆、印在微波电路板上的微带以及传送用于以极高速率传输数据的光波的光纤。

从根本上来说，传输线是一个双端口网络，每个端口由两个终端组成，如图2-1所示。其中一个端口（线路的发送端）连接到源（也称为发生器）。另一个端口，即线路的接收端，连接到负载。连接到传输线发送端的源可以是任何产生输出电压的电路，例如雷达发射器、放大器或在传输模式下运行的计算机终端。从电路理论来看，直流电源可以用 Thévenin 等效发电机电路来表示，该电路由发电机电压 V_g 与发电机电阻 R_g 串联组成，如图 2-1 所示。对于交流 (ac) 信号，发生器电路由电压相量 $\tilde{V}_g$ 和阻抗 Z_g 表示。

负载电路，或者简称为负载，可以是雷达的发射天线、在接收模式下运行的计算机终端、放大器的输入端子，或者其输入端子可以用等效负载阻抗 Z_L 表示的任何输出电路。

2-1.1 波长的作用

在低频电路中，电路元件通常使用简单的电线互连。例如，在图2-2所示的电路中，发电机通过一对电线连接到简单的RC负载。鉴于我们在前面几段中对传输线构成的定义，我们提出一个问题：端子 AA′ 和端子 BB′ 之间的一对电线是传输线吗？如果是这样，在什么情况下我们应该明确地将这对电线视为传输线，而不是完全忽略它们的存在并将电路视为仅连接到发电机 $\tilde{V}_g$ 的 RC 负载？第一个问题的答案是：是的，

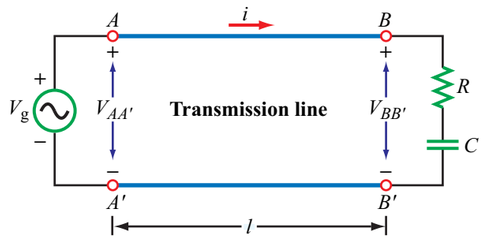


图 2-2 发电机通过长度为 l 的传输线连接到 RC 电路。

the pair of wires does constitute a transmission line. The answer to the second question is: *The factors that determine whether or not we should treat the wires as a transmission line is governed by the length of the line l and the frequency f of the signal provided by the generator*. (As we will see later, the determining factor is the ratio of the length l to the wavelength λ of the wave propagating on the transmission line between the source at terminals AA' and the load at terminals BB'.) If the generator voltage is cosinusoidal in time, then the voltage across the input terminals AA' is

$$V_{AA'} = V_g(t) = V_0 \cos \omega t \qquad \text{(V)}, \tag{2.1}$$

where $\omega = 2\pi f$ is the angular frequency. If we assume that the current flowing through the wires travels at the speed of light, $c = 3 \times 10^8$ m/s, then the voltage across the output terminals BB' will have to be delayed in time relative to that across AA' by the travel delay time l/c. Thus, assuming no ohmic losses in the transmission line and ignoring other transmission-line effects discussed later in this chapter,

$$V_{BB'}(t) = V_{AA'}(t - l/c) = V_0 \cos[\omega(t - l/c)] = V_0 \cos(\omega t - \phi_0), \tag{2.2}$$

with

$$\phi_0 = \frac{\omega l}{c} \qquad \text{(rad)}. \tag{2.3}$$

Thus, the time delay associated with the length of the line l manifests itself as a constant phase shift ϕ_0 in the argument of the cosine. Let us compare $V_{BB'}$ with $V_{AA'}$ at $t = 0$ for an ultralow-frequency electronic circuit operating at a frequency $f = 1$ kHz. For a typical wire length $l = 5$ cm, Eqs. (2.1) and (2.2) give $V_{AA'} = V_0$ and $V_{BB'} = V_0 \cos(2\pi f l/c) = 0.999999999998 V_0$. Hence, for all practical purposes, the presence of the transmission line may be ignored, and terminal AA' may be treated as identical with BB' so far as its voltage is concerned. On the other hand, had the line been a 20-km long telephone cable carrying a 1 kHz voice signal, then the same calculation would have led to $V_{BB'} = 0.91 V_0$, which is a deviation of 9%. Furthermore, had the frequency of the signal on the 5-cm long wire been 1.5 GHz instead of 1 kHz, the voltage at the end of the line would have been $V_{BB'} = 0$! Thus, the pair of wires would have a voltage V_0 at AA' and zero at BB' simultaneously. The determining factor is the magnitude of $\phi_0 = \omega l/c$. From Eq. (1.27), the velocity of propagation u_p of a traveling wave is related to the oscillation frequency f and the wavelength λ by

$$u_p = f\lambda \qquad \text{(m/s)}.$$

In the present case, $u_p = c$. Hence, the phase delay is

$$\phi_0 = \frac{\omega l}{c} = \frac{2\pi f l}{c} = 2\pi \frac{l}{\lambda} \quad \text{radians}. \tag{2.4}$$

▶ When l/λ is very small, transmission-line effects may be ignored, but when $l/\lambda \gtrsim 0.01$, it may be necessary to account not only for the phase shift due to the time delay but also for the presence of ***reflected*** signals that may have been bounced back by the load toward the generator. ◀

Power loss on the line and ***dispersive*** effects may need to be considered as well.

▶ A dispersive transmission line is one on which the wave velocity is not constant as a function of the frequency f. ◀

This means that the shape of a rectangular pulse, which through Fourier analysis can be decomposed into many sinusoidal waves of different frequencies, gets distorted as it travels down the line because its different frequency components do not propagate at the same velocity (**Fig. 2-3**). Preservation of pulse shape is very important in high-speed data transmission not only between terminals but also across transmission line segments fabricated within high-speed integrated circuits. At 10 GHz, for example, the wavelength is $\lambda = 3$ cm in air but only on the order of 1 cm in a semiconductor material. Hence, even lengths between devices on the order of millimeters become significant, and their presence has to be accounted for in the design of the circuit.

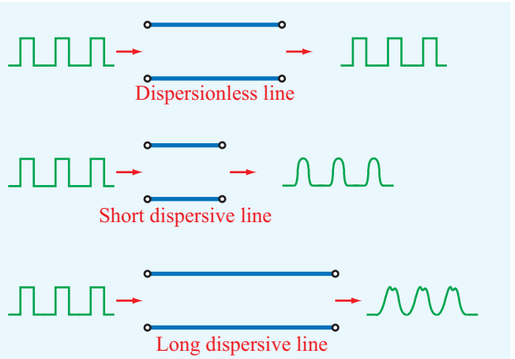


Figure 2-3 A dispersionless line does not distort signals passing through it regardless of its length, whereas a dispersive line distorts the shape of the input pulses because the different frequency components propagate at different velocities. The degree of distortion is proportional to the length of the dispersive line.

这对电线确实构成了传输线。第二个问题的答案是：决定我们是否应该将电线视为传输线的因素取决于线的长度l和发生器提供的信号的频率f。（正如我们稍后将看到的，决定因素是在端子 AA′ 处的源和端子 BB′ 处的负载之间的传输线上传播的波的长度 l 与波长 λ 的比率。）如果发电机电压随时间呈余弦曲线，则输入端子 AA′ 两端的电压为

$$V_{AA'} = V_g(t) = V_0 \cos \omega t \qquad \text{(V)}, \tag{2.1}$$

其中 $\omega = 2\pi f$ 是角频率。如果我们假设流过导线的电流以光速 $c = 3 \times 10^8$ m/s 传播，那么输出端子 BB′ 两端的电压将必须相对于 AA′ 两端的电压在时间上延迟 l/c。因此，假设传输线中没有欧姆损耗并忽略本章后面讨论的其他传输线效应，

$$V_{BB'}(t) = V_{AA'}(t - l/c) = V_0 \cos[\omega(t - l/c)] = V_0 \cos(\omega t - \phi_0), \tag{2.2}$$

和

$$\phi_0 = \frac{\omega l}{c} \qquad \text{(rad)}. \tag{2.3}$$

因此，与线 l 的长度相关的时间延迟表现为余弦参数中的恒定相移 ϕ_0。让我们比较一下 $t = 0$ 时的 $V_{BB'}$ 与 $V_{AA'}$，了解以频率 $f = 1$ kHz 运行的超低频电子电路。对于典型的线长 $l = 5$ cm，方程：(2.1) 和 (2.2) 给出 $V_{AA'} = V_0$ 和 $V_{BB'} = V_0 \cos(2\pi f l/c) = 0.999999999998 V_0$。因此，出于所有实际目的，可以忽略传输线的存在，并且就其电压而言，端子AA′可以被视为与BB′相同。另一方面，如果线路是传输 1 kHz 语音信号的 20 公里长的电话电缆，那么相同的计算将导致 $V_{BB'} = 0.91 V_0$，偏差为 9%。此外，如果 5 厘米长的电线上的信号频率为 1.5 GHz 而不是 1 kHz，则线路末端的电压将为 $V_{BB'} = 0$！因此，这对导线在 AA′ 处具有电压 V_0，在 BB′ 处同时具有零电压。决定因素是$\phi_0 = \omega l/c$ 的大小。从方程。根据方程 (1.27)，行波的传播速度 u_p 与振荡频率 f 和波长 λ 的关系为：

$$u_p = f\lambda \qquad \text{(m/s)}.$$

在本例中，$u_p = c$。因此，相位延迟为

$$\phi_0 = \frac{\omega l}{c} = \frac{2\pi f l}{c} = 2\pi \frac{l}{\lambda} \quad \text{radians}. \tag{2.4}$$

▶ 当 l/λ 非常小时，可以忽略传输线效应，但当 $l/\lambda \gtrsim 0.01$ 时，可能不仅需要考虑由于时间延迟而导致的相移，还需要考虑可能已被负载反射回发电机的反射信号的存在。◀

线路上的功率损耗和色散效应也可能需要考虑。

▶ 色散传输线是一种波速作为频率 f 的函数并不恒定的传输线。◀

这意味着矩形脉冲的形状（通过傅里叶分析可以分解为许多不同频率的正弦波）在沿线路传播时会发生畸变，因为其不同频率分量不以相同的速度传播（图2-3）。脉冲形状的保持在高速数据传输中非常重要，不仅在终端之间，而且在高速集成电路内制造的传输线段之间也是如此。例如，在 10 GHz 时，空气中的波长为 $\lambda = 3$ cm，但在半导体材料中仅为 1 cm 左右。因此，即使器件之间的长度为毫米级，也变得很重要，并且在电路设计中必须考虑到它们的存在。

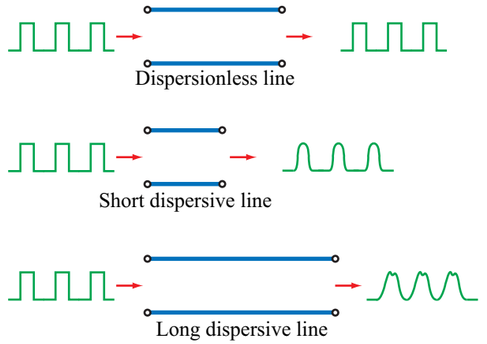


图 2-3 无色散线无论其长度如何都不会扭曲通过它的信号，而色散线会扭曲输入脉冲的形状，因为不同的频率分量以不同的速度传播。畸变程度与色散线的长度成正比。

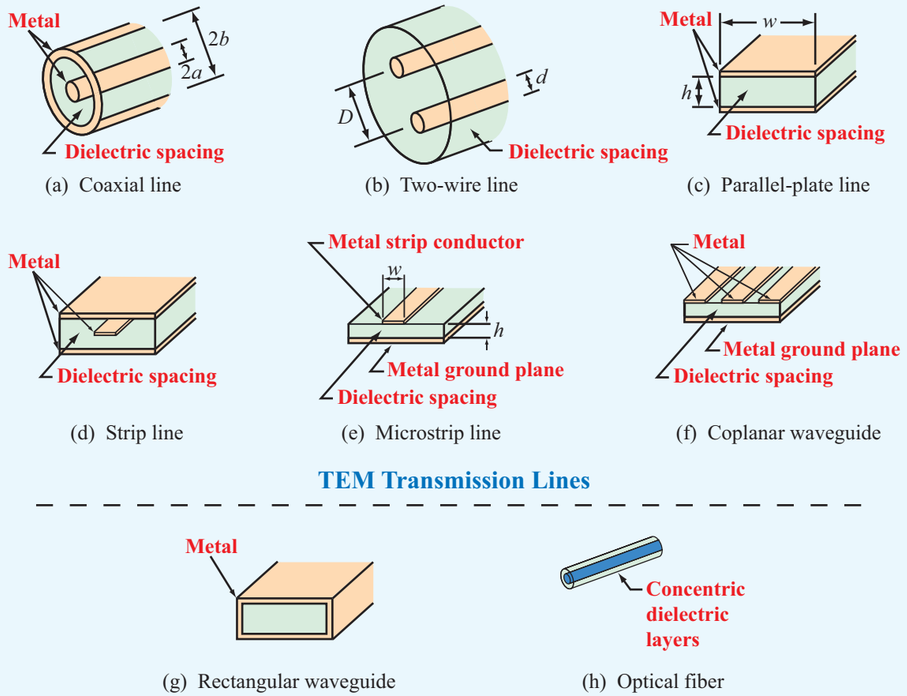


Figure 2-4 A few examples of transverse electromagnetic (TEM) and higher-order transmission lines.

2-1.2 Propagation Modes

A few examples of common types of transmission lines are shown in **Fig. 2-4**. Transmission lines may be classified into two basic types:

- ***Transverse electromagnetic* (*TEM*) *transmission lines:*** Waves propagating along these lines are characterized by electric and magnetic fields that are entirely ***transverse*** to the direction of propagation. Such an orthogonal configuration is called a ***TEM mode***. A good example is the coaxial line shown in **Fig. 2-5**. The electric field is in the radial direction between the inner and outer conductors, while the magnetic field circles the inner conductor, and neither has a component along the line axis (the direction of wave propagation). Other TEM transmission lines include the two-wire line and the parallel-plate line (both shown in **Fig. 2-4**). Although the fields present on a microstrip line do not adhere to the exact definition of a TEM mode, the nontransverse field components are sufficiently small (in comparison with the transverse components) that they may be ignored, thereby allowing the inclusion of microstrip lines in the TEM class. A common feature among TEM lines is that they consist of two parallel conducting surfaces.

- ***Higher-order transmission lines:*** Waves propagating along these lines have at least one significant field component in the direction of propagation. Hollow conducting waveguides, dielectric rods, and optical fibers belong to this class of lines (Chapter 8).

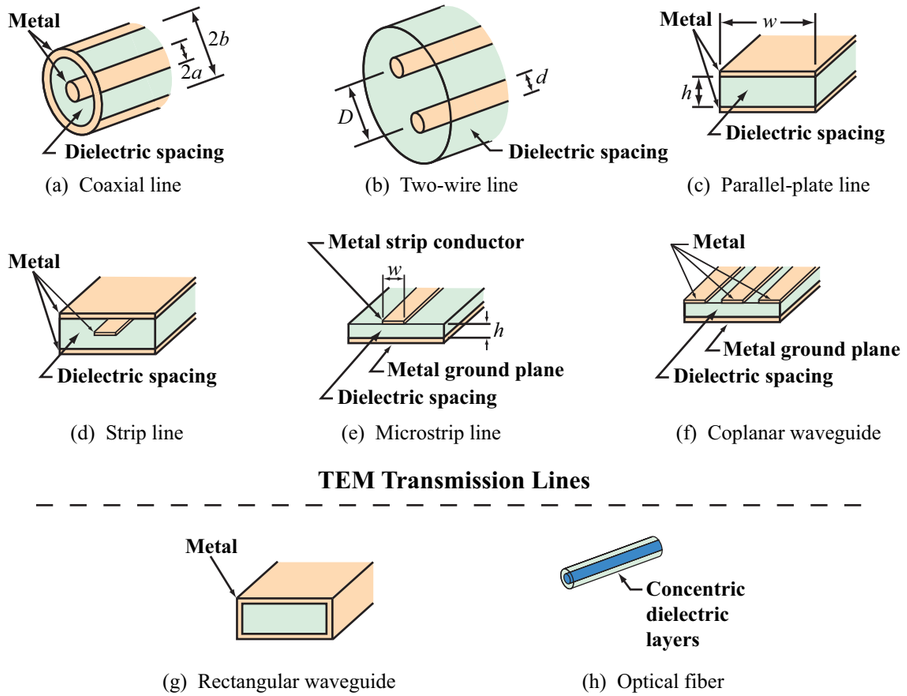


图 2-4 横向电磁 (TEM) 和高阶传输线的几个示例。

2-1.2 传播模式

图 2-4 显示了一些常见类型的传输线示例。传输线可分为两种基本类型：

- 横向电磁 (TEM) 传输线：沿这些线传播的波的特点是电场和磁场完全横向于传播方向。这种正交配置称为 TEM 模式。图 2-5 所示的同轴线就是一个很好的例子。电场沿内导体和外导体之间的径向方向，而磁场围绕内导体，并且两者都没有沿线轴（波传播方向）的分量。其他TEM传输线包括双线线路和平行板线路（均如图2-4所示）。尽管微带线上的场不符合 TEM 模式的精确定义，但非横向场分量足够小（与横向分量相比），可以忽略它们，从而允许将微带线包含在 TEM 类中。 TEM 线的一个共同特征是它们由两个平行的导电表面组成。

- 高阶传输线：沿这些线传播的波在传播方向上至少有一个重要的场分量。空心传导波导、介电棒和光纤都属于此类线路（第 8 章）。

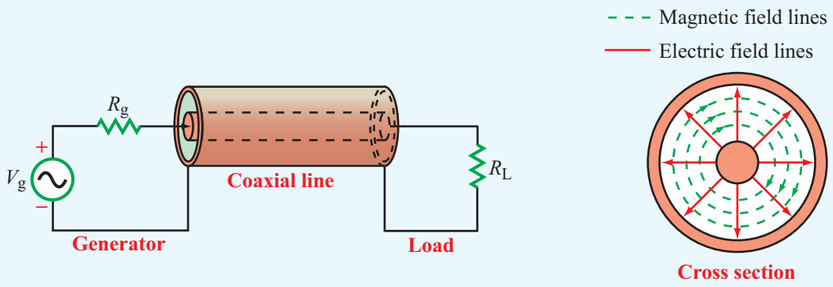


Figure 2-5 In a coaxial line, the electric field is in the radial direction between the inner and outer conductors, and the magnetic field forms circles around the inner conductor. The coaxial line is a transverse electromagnetic (TEM) transmission line because both the electric and magnetic fields are orthogonal to the direction of propagation between the generator and the load.

Only TEM-mode transmission lines are treated in this chapter. This is because they are more commonly used in practice, and fortunately, less mathematical rigor is required for treating them than is required for lines that support higher-order modes. We start our treatment by representing the transmission line in terms of a lumped-element circuit model, and then we apply Kirchhoff's voltage and current laws to derive a pair of equations governing their behavior, known as the ***telegrapher's equations***. By combining these equations, we obtain wave equations for the voltage and current at any location along the line. Solution of the wave equations for the sinusoidal steady-state case leads to a set of formulas that can be used to solve a wide range of practical problems. In the latter part of this chapter, we introduce a graphical tool known as the ***Smith chart***, which facilitates the solution of transmission-line problems without having to perform laborious calculations involving complex numbers.

2-2 Lumped-Element Model

When we draw a schematic of an electronic circuit, we use specific symbols to represent resistors, capacitors, inductors, diodes, and the like. In each case, the symbol represents the functionality of the device rather than its shape, size, or other attributes. We shall do the same for transmission lines.

► A transmission line will be represented by a parallel-wire configuration (**Fig. 2-6(a)**), regardless of its specific shape or constitutive parameters. ◄

Thus, **Fig. 2-6(a)** may represent a coaxial line, a two-wire line, or any other TEM line.

Drawing again on our familiarity with electronic circuits, when we analyze a circuit containing a transistor, we mimic the functionality of the transistor by an equivalent circuit composed of sources, resistors, and capacitors. We apply the same approach to the transmission line by orienting the line along the z direction, subdividing it into differential sections each of length Δz (**Fig. 2-6(b)**), and then representing each section by an equivalent circuit, as illustrated in **Fig. 2-6(c)**. This representation, often called the ***lumped-element circuit*** model, consists of four basic elements with values that henceforth will be called the ***transmission line parameters***. These are:

- R': The combined ***resistance*** of both conductors per unit length, in Ω/m,
- L': The combined ***inductance*** of both conductors per unit length, in H/m,
- G': The ***conductance*** of the insulation medium between the two conductors per unit length, in S/m, and
- C': The ***capacitance*** of the two conductors per unit length, in F/m.

Whereas the four line parameters are characterized by different formulas for different types of transmission lines, the equivalent model represented by **Fig. 2-6(c)** is equally applicable to all TEM transmission lines. *The prime superscript is used as a reminder that the line parameters are differential quantities whose units are per unit length.*

Expressions for the line parameters R', L', G', and C' are given in **Table 2-1** for the three types of TEM transmission lines diagrammed in parts (a) through (c) of **Fig. 2-4**. For

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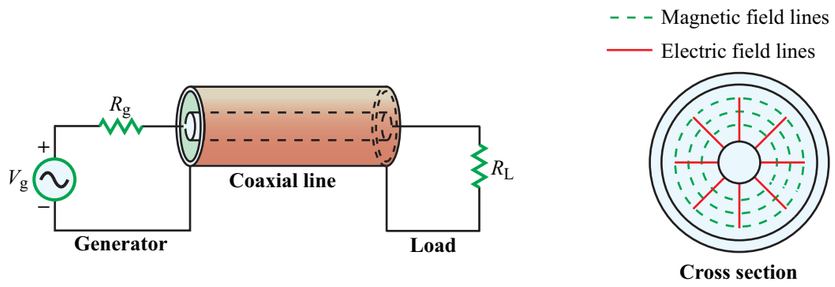


图2-5 在同轴线中，电场沿内导体和外导体之间的径向方向，磁场围绕内导体形成圆圈。同轴线是横向电磁 (TEM) 传输线，因为电场和磁场都与发电机和负载之间的传播方向正交。

本章仅讨论 TEM 模式传输线。这是因为它们在实践中更常用，幸运的是，处理它们所需的数学严谨性比支持高阶模式的线所需的数学严谨性要少。我们首先用集总元件电路模型表示传输线，然后应用基尔霍夫电压和电流定律推导出一对控制其行为的方程，称为电报员方程。通过组合这些方程，我们可以获得沿线任意位置的电压和电流的波动方程。正弦稳态情况下波动方程的求解得出了一组可用于解决各种实际问题的公式。在本章的后半部分，我们将介绍一种称为史密斯圆图的图形工具，它有助于解决传输线问题，而无需执行涉及复数的繁琐计算。

2-2 集总元件模型

当我们绘制电子电路原理图时，我们使用特定的符号来表示电阻器、电容器、电感器、二极管等。在每种情况下，符号都代表设备的功能，而不是其形状、大小或其他属性。我们将对传输线做同样的事情。

► 传输线将由平行线配置表示（图2-6（a）），无论其具体形状或本构参数如何。 ◄

因此，图2-6(a)可以表示同轴线、两线或任何其他TEM线。

再次利用我们对电子电路的熟悉，当我们分析包含晶体管的电路时，我们通过由源、电阻器和电容器组成的等效电路来模拟晶体管的功能。我们将相同的方法应用于传输线，将传输线沿 z 方向定向，将其细分为每个长度为 Δz 的差分部分（图 2-6（b））， 然后用等效电路表示每个部分，如图 2-6（c）所示。这种表示法通常称为集总元件电路模型，由四个基本元件组成，其值今后将称为传输线参数。这些都是：

- R′：单位长度两个导体的组合电阻，单位为Ω/m，
- L′：每单位长度两个导体的组合电感，单位为 H/m，
- G′：单位长度两导体之间绝缘介质的电导率，单位为S/m，
- C ′：单位长度上两个导体的电容，单位为F/m。

尽管对于不同类型的传输线，四个线参数有不同的公式来表征，但图 2-6（c）所示的等效模型同样适用于所有 TEM 传输线。主上标用于提醒，线参数是微分量，其单位为每单位长度。

对于图2-4(a)至(c)中所示的三种类型的TEM传输线，表2-1给出了线参数R′、L′、G′和C′的表达式。为了